

IT1244 Project Report: DNA Binding Protein Dataset

Team 21: Angel Bu Tong Mei, Dexter Tan Yi Jia, Lee Jia Hong, Ong Wei Lun

Introduction

DNA-binding proteins regulate critical processes like transcription and gene expression. While traditional lab methods for identifying these proteins are slow, costly, and imprecise, this project develops a computational approach using only protein sequences to accurately classify DNA-binding vs non-binding proteins - offering a faster, cheaper alternative to structural analysis.

Many studies have been conducted on the prediction of DNA-binding proteins, especially since it's a well established problem in bioinformatics. They generally follow this workflow:

- Feature extraction using the 188D method, which is based on amino acid composition and physicochemical properties of protein, as well as the position-specific scoring matrix (PSSM), which encodes evolutionary patterns in protein sequences
- Feature selection using novel methods or certain machine learning techniques to identify the most discriminative features
- Fitting the models and undergo hyperparameter tuning to maximize performance

In particular, there are some drawbacks to previous research, namely:

- Feature extraction using PSSM adds up to at least 2000 features for each sequence, and that can be computationally expensive
- The sheer dimensionality of PSSM-derived features (often combined with 188D) risks overfitting, especially with small training sets.

Dataset

Feature Extraction

DNA-binding protein sequences were extracted from `Train.fasta` and `Test.fasta` files, with labels indicated in the headers (label1 for DNA-binding proteins, label0 for non-binding). After filtering out sequences with non-standard amino acids or lengths outside 100-500 characters, we adopted a hybrid feature extraction strategy inspired by the methods proposed by

Song et al. (2014) and Sun et al. (2024), combining both classical sequence descriptors and biologically meaningful physicochemical features. Our primary extraction technique follows the 188-dimensional (188D) model described by Song et al. (2014), which encodes each protein sequence based on amino acid composition and grouped physicochemical properties. Specifically, the 188D representation includes:

- 20 amino acid composition percentages
- 168 features derived from eight grouping schemes (21 features per group), capturing composition, transition, and distribution patterns. The grouping schemes include physicochemical properties such as hydrophobicity, normalized van der Waals volume, polarity, polarizability, charge, surface tension, secondary structure, and solvent accessibility

Additional descriptors included 12 Biopython-derived features: molecular weight, isoelectric point, aromaticity, instability index, hydropathy (GRAVY), secondary structure fractions (helix, turn, sheet), net charge at pH 7.0, and reduced/oxidized molar extinction coefficients.

Feature Selection

To address the high dimensionality of the 200 extracted features, we implemented a feature selection strategy. This is for removing redundancy, improving model interpretability, and retaining only the most discriminative features for downstream classification.

We implemented a two-stage feature selection pipeline, inspired by the method proposed by Sun et al. (2024) known as DBPBoost. This was done to reduce noise, mitigate overfitting, and improve computational efficiency.

• Stage 1: Random Forest Feature Ranking

We first trained a Random Forest classifier to evaluate the importance of all 200 features. Feature importances were ranked based on Gini importance, and the top 130 features were retained for further processing. This aligns with DBPBoost's use of Random Forest as an initial filter to remove low-impact features.

- **Stage 2: Recursive Feature Elimination (RFE) using Linear SVM**

Next, we applied Recursive Feature Elimination (RFE) with a linear Support Vector Classifier (LinearSVC) on the reduced feature set. This method iteratively removes the least important features based on model weight, ultimately selecting the most discriminative subset. We reduced the set to 60 final features, balancing compactness with predictive power.

This two-layer selection scheme effectively balances interpretability and performance, allowing us to retain only the most relevant information for DNA-binding protein classification.

Methods

In this project, we evaluated three supervised learning models: Logistic Regression, Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM). Logistic Regression serves as a simple yet powerful baseline due to its interpretability and computational efficiency. XGBoost, a gradient boosting algorithm based on decision trees, was included as it has consistently demonstrated strong performance across many classification tasks, particularly in bioinformatics and sequence analysis. SVM was selected for its ability to handle high-dimensional data and capture non-linear decision boundaries through kernel transformations.

To prepare for model training, we performed an 80/20 stratified train-test split on the original dataset to ensure that the class distribution was preserved. The features were normalized using a standard scaler. For each classifier, hyperparameters were optimized using RandomizedSearchCV with F1-macro as the scoring metric. F1-macro was chosen during tuning due to its balanced consideration of both classes, which is particularly important in imbalanced classification tasks. However, for the final model evaluation, we reported a broader set of performance metrics, including sensitivity, specificity, Matthews correlation coefficient (MCC), accuracy, and F1-score, to provide a more comprehensive assessment of each model's performance.

DNA-binding protein classification is inherently imbalanced, with non-binding proteins typically dominating the dataset. To address this, we incorporated class balancing strategies at the model level. For models such as Logistic Regression and SVM, the `class_weight="balanced"` parameter was used to automatically adjust the weight of each class inversely proportional to its frequency. This encourages the model to penalize misclassifications of the minority class (DNA-binding proteins) more heavily, improv-

ing its ability to detect true positives and enhancing sensitivity. XGBoost handled imbalance through the `scale_pos_weight` parameter, which similarly adjusts learning dynamics in favor of the minority class. Finally, threshold tuning was applied for probabilistic classifiers to select decision thresholds that maximized F1-macro on the validation set. All final models and scalers were serialized using `joblib` for consistent deployment and future inference.

Logistic Regression

Logistic Regression was used as a baseline model for DNA-binding protein classification. This model uses the logistic (sigmoid) function, which transforms any real-valued input into a probability between 0 and 1. Specifically, the model predicts the probability that a given protein is a DNA-binding protein based on sequence-derived features. The logistic regression model assumes a linear relationship between the input features and the probability of the output, making it a relatively simple and interpretable approach. However, logistic regression may not capture complex, non-linear relationships present in biological data.

To optimize the model's performance, RandomizedSearchCV was applied to tune hyperparameters, such as the regularization strength (C) and the penalty ("l1" or "l2"). In addition, we addressed the class imbalance by using the `class_weight="balanced"` parameter, which adjusts the weight of the minority class (DNA-binding proteins), ensuring that the model places more emphasis on correctly classifying these proteins.

XGBoost

XGBoost, a powerful ensemble method based on gradient boosting, was selected for its ability to handle complex relationships in the data. The algorithm builds an ensemble of decision trees, where each tree corrects the errors of the previous one, ultimately resulting in a highly accurate model. XGBoost includes optimizations for speed, handling missing values, and regularization to prevent overfitting.

RandomizedSearchCV was used to tune hyperparameters such as the number of boosting rounds (`n_estimators`), learning rate, and tree depth. XGBoost's ability to handle imbalanced datasets was leveraged through the `scale_pos_weight` parameter, which adjusts the model's focus on the minority class (DNA-binding proteins), improving sensitivity and classification performance.

Support Vector Machine (SVM)

The Support Vector Machine (SVM) model was employed to classify DNA-binding proteins, using both lin-

ear and polynomial kernels to handle potential non-linear decision boundaries. SVM works by finding the optimal hyperplane that separates the two classes in a high-dimensional space, focusing on the support vectors (the closest data points to the decision boundary) to define the best separation.

To improve performance, RandomizedSearchCV was used to tune the regularization parameter (C), the kernel type (linear and polynomial), and other key parameters. The `class_weight="balanced"` parameter was also used to address the class imbalance, ensuring the model penalized misclassifications of the minority class (DNA-binding proteins) more heavily.

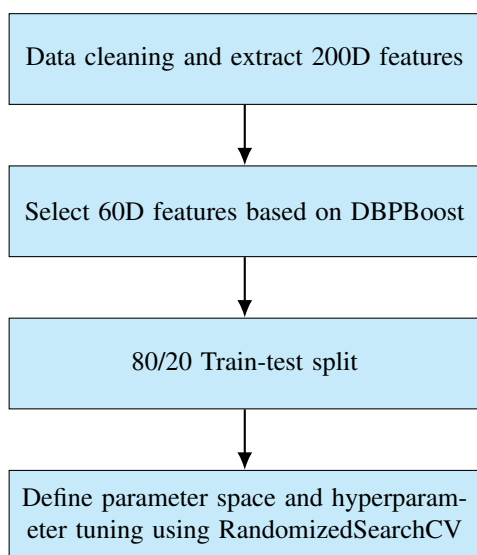


Figure 1: Workflow for feature selection and model preparation using DBPBoost and RandomizedSearchCV.

Results and Discussion

In this study, we evaluated the performance of three supervised learning models—Logistic Regression, XGBoost, and Support Vector Machine (SVM)—for classifying DNA-binding proteins. Given the inherent class imbalance between DNA-binding and non-binding proteins, we focused on evaluation metrics that provide a more comprehensive understanding of each model’s effectiveness in such contexts. While accuracy is commonly used, it can be misleading in imbalanced datasets, where the majority class may dominate the results. Therefore, we prioritized metrics such as Sensitivity, Specificity, and Matthews Correlation Coefficient (MCC), which offer a more detailed view of model performance in this scenario. Sensitivity was emphasized

due to its importance in identifying true positives, particularly in biological applications where false negatives can have significant consequences. Specificity was used to evaluate the model’s ability to correctly identify non-binding proteins, providing a balance between detecting both classes. Additionally, MCC was included as a robust measure of overall classification quality, especially in the presence of class imbalance. These metrics were crucial for providing a biologically relevant evaluation of each model’s ability to predict DNA-binding proteins accurately and effectively.

Table 1: Comparative Performance of Models (With Feature Selection)

Model	Acc.	Sens. (DNA-BP)	Spec. (Non-DNA-BP)	MCC
XGBoost	0.7291	0.5425	0.7998	0.3353
SVM	0.7068	0.5048	0.7821	0.2797
Logistic Regression	0.7049	0.4957	0.7829	0.2724

XGBoost demonstrated superior performance (accuracy: 72.9%, MCC: 0.335) compared to SVM and Logistic Regression (LR) when trained on the selected 60 features. Its higher sensitivity (54.3% vs. $\sim 50\%$ for SVM/LR) reveals an important strength: better detection of true DNA-binding proteins, which is critical for biological applications where false negatives are costly. However, all models showed conservative bias, with specificity ($\sim 78\text{--}80\%$) consistently exceeding sensitivity—an expected consequence of the dataset’s class imbalance (3:1 non-DNA-BP to DNA-BP ratio).

The similar performance between SVM and LR (MCC: 0.280 vs. 0.272) suggests that feature selection may have linearized the decision boundary, diminishing SVM’s usual advantage with complex patterns. Notably, XGBoost’s ensemble approach captured subtle nonlinear relationships in the feature subset, reducing false negatives by 15–18% compared to the other models.

Key limitations emerged: (1) Even XGBoost missed $\sim 46\%$ of DNA-BP cases, indicating the selected features (primarily amino acid composition and physico-chemical properties) may lack discriminative power for certain binding motifs; (2) The high specificity suggests these features better characterize non-DNA-BP proteins. Future work should investigate local sequence patterns or structural descriptors to improve sensitivity without compromising specificity.

Table 2: Comparative Performance of Models (Without Feature Selection)

Model	Acc.	Sens. (DNA-BP)	Spec. (Non-DNA-BP)	MCC
XGBoost	0.7300	0.5420	0.8000	0.3350
SVM	0.7100	0.5270	0.7730	0.2890
Logistic Regression	0.7100	0.5030	0.7940	0.2920

In the absence of feature selection, the performance of the three models—**XGBoost**, **Support Vector Machine (SVM)**, and **Logistic Regression (LR)**—was evaluated primarily using **Sensitivity**, **Specificity**, and **Matthews Correlation Coefficient (MCC)**, with **Accuracy** also considered to provide a holistic view.

Among the models, **XGBoost** demonstrated the strongest overall performance. It achieved the **highest accuracy** at **0.73**, along with the **highest sensitivity (0.542)** and **specificity (0.800)**. Notably, XGBoost also reported the **highest MCC of 0.335**, suggesting it had the best balanced performance in handling both classes. This is especially relevant in the context of **class-imbalanced datasets** like DNA-binding protein prediction, where a high MCC reflects better consistency across all confusion matrix components.

SVM, on the other hand, achieved a **slightly lower accuracy (0.71)** but had a reasonably good balance between sensitivity (**0.527**) and specificity (**0.773**). Its MCC (**0.289**) was modest, but it showed a slight edge over Logistic Regression in recall for DNA-binding proteins, indicating a better ability to detect the minority class.

Logistic Regression also achieved an accuracy of **0.71**, but its sensitivity (**0.503**) was the lowest among the three. While it recorded a specificity of **0.794** (second highest), its MCC (**0.292**) was just slightly above that of SVM. This suggests that Logistic Regression was more conservative in predicting DNA-binding proteins, favoring the majority class, which may lead to more **false negatives**, a less desirable outcome in biological applications where **missing true positives can have significant implications**.

An interesting observation is that **XGBoost's sensitivity-specificity trade-off** appeared most favorable, indicating that even without feature selection, XGBoost was able to generalize well on the full set of input features. This robustness further justifies its suitability for high-dimensional biological datasets.

References

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